



2014 TRIAL HIGHER SCHOOL CERTIFICATE

PM Thursday 1st August

Section I – Multiple Choice

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

 (A) (B) (C) (D)

(A)  (B)  (C)  (D) 

(A)  (B)  *correct* (C)  (D) 

Start Here →

- | | | | | |
|-----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1. | A ☐ | B ☐ | C ☐ | D ☐ |
| 2. | A ☐ | B ☐ | C ☐ | D ☐ |
| 3. | A ☐ | B ☐ | C ☐ | D ☐ |
| 4. | A ☐ | B ☐ | C ☐ | D ☐ |
| 5. | A ☐ | B ☐ | C ☐ | D ☐ |
| 6. | A ☐ | B ☐ | C ☐ | D ☐ |
| 7. | A ☐ | B ☐ | C ☐ | D ☐ |
| 8. | A ☐ | B ☐ | C ☐ | D ☐ |
| 9. | A ☐ | B ☐ | C ☐ | D ☐ |
| 10. | A ☐ | B ☐ | C ☐ | D ☐ |

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Barker College

Student Number:

2014
YEAR 12
TRIAL HSC
EXAMINATION

Mathematics Extension 2

Staff Involved:

PM 1st AUGUST 2014

- BHC
- VAB
- KJL*
- GDH*

Number of copies: 40

General Instructions

- Reading time - 5 minutes
- Working time - 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2-6

10 marks

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II Pages 7-15

90 marks

- Attempt Questions 11 - 16
- Allow about 2 hours 45 minutes for this section

Section I — Multiple Choice

10 marks

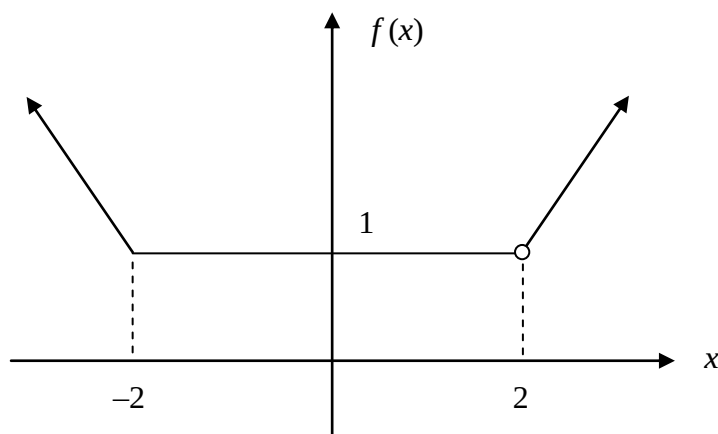
Attempt Questions 1-10. Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-10.

1. What is the eccentricity of the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$?

- (A) $\frac{2}{3}$
- (B) $\frac{3}{2}$
- (C) $\frac{\sqrt{5}}{3}$
- (D) $\frac{3}{\sqrt{5}}$

2. The graph of a function $y = f(x)$ is shown below.



Which of the following statements is true?

- (A) $f(x)$ is not continuous at $x = 2$ and $f(x)$ is not differentiable for $-2 < x < 2$
- (B) $f(x)$ is not continuous at $x = 2$ and $x = -2$, and $f(x)$ is not differentiable at $x = 2$ and $x = -2$
- (C) $f(x)$ is not continuous at $x = 2$, but $f(x)$ is differentiable for all x
- (D) $f(x)$ is not continuous at $x = 2$ and not differentiable at $x = 2$ and $x = -2$

Section I continued

3. The graph below shows a hyperbola, including the equations of the directrices and the coordinates of the foci.

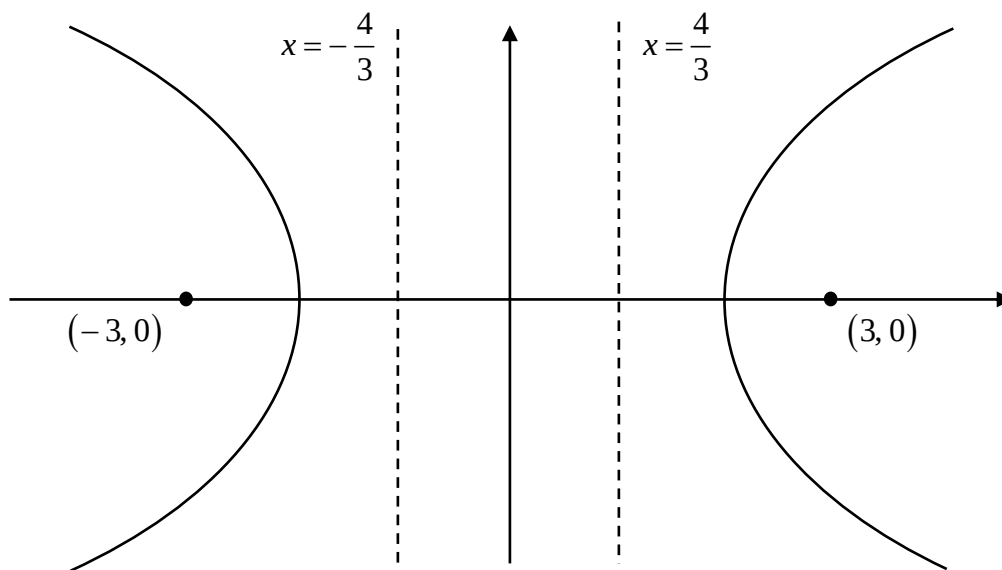


DIAGRAM NOT TO SCALE

The equation of this hyperbola is:

- (A) $\frac{x^2}{4} - \frac{y^2}{5} = 1$
- (B) $\frac{x^2}{5} - \frac{y^2}{4} = 1$
- (C) $\frac{x^2}{16} - \frac{y^2}{25} = 1$
- (D) $\frac{x^2}{2} - \frac{y^2}{5} = 1$
4. The polynomial $P(x) = x^4 + mx^2 + nx + 28$ has a double root at $x = 2$.

What are the values of m and n ?

- (A) $m = -11$ and $n = -12$
- (B) $m = -5$ and $n = -12$
- (C) $m = -11$ and $n = 12$
- (D) $m = -5$ and $n = 12$

Section I continued

5. Diagram A shows the complex number z represented in the Argand plane.

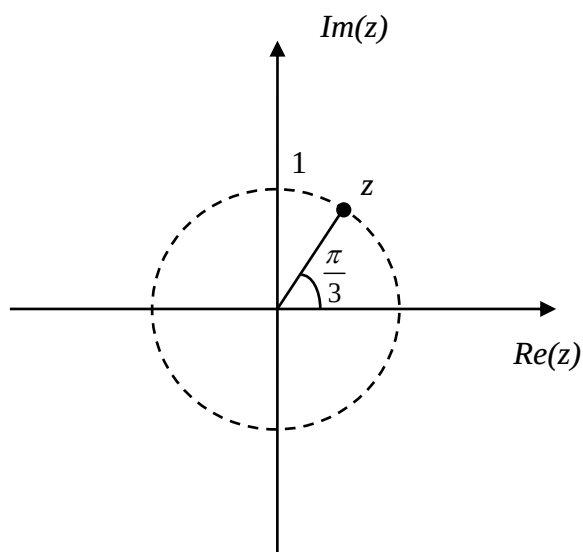


DIAGRAM A

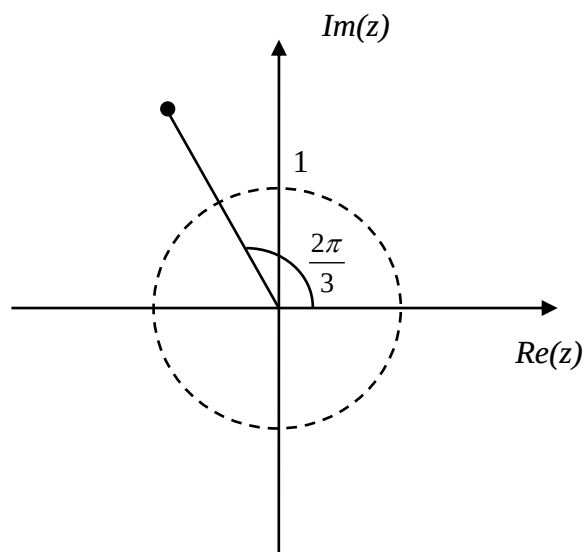


DIAGRAM B

Diagram B shows:

- (A) z^2
- (B) $2iz$
- (C) $-2z$
- (D) $2z^2$
6. Let α, β, γ be the roots of $2x^3 - 5x^2 + 2x - 1 = 0$.
The equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$ is best given by:

- (A) $x^3 + 2x^2 + 25x - 2 = 0$
- (B) $x^3 - 2x^2 + 5x - 2 = 0$
- (C) $x^3 - 2x^2 + 25x - 8 = 0$
- (D) $-2x^3 + 5x^2 - 2x + 1 = 0$

Section I continued

7. How many distinct horizontal tangents can be drawn on the graph of $x^2 + y^2 + xy - 5 = 0$?

- (A) 0
- (B) 1
- (C) 2
- (D) More than 2

8. $\int_0^{\frac{\pi}{4}} \tan^3 x \, dx$ is equal in value to:

(A) $\frac{1}{2} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \tan^3 x \, dx$

(B) $\int_0^{\frac{\pi}{4}} \frac{\pi}{4} - \tan^3 x \, dx$

(C) $\int_0^{\frac{\pi}{4}} \tan x (\sec^2 x + 1) \, dx$

(D) $\int_0^{\frac{\pi}{4}} \left(\frac{1 - \tan x}{1 + \tan x} \right)^3 dx$

Section I continued

9. The value of $\int_{-2}^{-1} \sqrt{4-x^2} \, dx + \int_1^2 \sqrt{4-x^2} \, dx$ is:

(A) $\frac{4\pi}{3} - \sqrt{3}$

(B) $\frac{2\pi}{3} - \sqrt{3}$

(C) $\frac{2\pi}{3} - 2\sqrt{3}$

(D) $\frac{8\pi}{3} - \sqrt{3}$

10. The statement $\int_0^1 \frac{dx}{1+x^n} < \int_0^1 \frac{dx}{1+x^{n+1}}$ is:

(A) Always true

(B) Always false

(C) Only true for positive n

(D) Only true for negative n

End of Section I

Section II

90 marks

Attempt Questions 11-16.

Allow about 2 hours 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Show relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)	[Use a SEPARATE writing booklet]	Marks
(a) (i)	Find the square root of the complex number $15 - 8i$.	2
(ii)	Hence or otherwise, solve $z^2 + 6z - 6 + 8i = 0$.	1
(b) (i)	Write the complex number $z = 1 + \sqrt{3}i$ in modulus-argument form.	1
(ii)	For what values of n is $(1 + \sqrt{3}i)^n$ completely real?	
	Justify your answer with appropriate reasoning.	2
(c)	The points A and B represent the complex numbers $z_1 = 2 - i$ and $z_2 = 8 + i$ respectively. Find all possible complex numbers z_3 , represented by C such that $\triangle ABC$ is isosceles and right-angled at C .	3
(d)	On the Argand diagram, sketch the locus of z defined by $\arg\left(\frac{z - (3 + 3i)}{z - 2}\right) = \frac{\pi}{3}$.	2
(e) (i)	Sketch $y = x(2^x)$, showing all key features.	2
(ii)	For what values of k does $x(2^x) - kx = 0$ have exactly 2 real roots?	2

End of Question 11

Question 12 (15 marks)**[Use a SEPARATE writing booklet]****Marks**

(a) (i) Find real numbers A , B and C , such that $\frac{A}{x-2} + \frac{Bx+C}{x^2+4} = \frac{1}{(x-2)(x^2+4)}$ **2**

(ii) Hence, or otherwise, find $\int \frac{1}{(x-2)(x^2+4)} dx$ **2**

(b) Using the substitution $t = \tan \frac{\theta}{2}$ or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{\sin \theta + 2}$ **3**

(c) By considering $y = x^2 - 4x + 3$, draw on separate diagrams sketches of:

(i) $y^2 = x^2 - 4x + 3$ **2**

(ii) $y = \frac{1}{x^2} - \frac{4}{x} + 3$ *Hint: You may wish to consider this graph as $y = f\left(\frac{1}{x}\right)$* **3**

(d) Let $I_n = \int x^n e^x dx$.

(i) Show that $I_n = x^n e^x - nI_{n-1}$. **1**

(ii) Hence evaluate $I = \int_1^2 x^2 e^x dx$ **2**

End of Question 12

Question 13 (15 marks)**[Use a SEPARATE writing booklet]****Marks**

- (a) The base of a solid is the region bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Its volume consists of semicircular cross sections.

Mr Lee initially believed that the volume of the solid would be the same regardless of whether the semicircular cross-sections were parallel to the x -axis or the y -axis.

Mr Lee was **wrong**.

By showing appropriate calculations, find the ratio of the volumes of the two solids.

4

- (b) Let $f(x) = 5$ and $g(x) = x + \frac{4}{x}$.

- (i) On the same diagram, sketch $y = f(x)$ and $y = g(x)$.

Clearly label any points of intersection.

2

- (ii) The region bounded by $y = f(x)$ and $y = g(x)$ is rotated about the line $x = -1$.

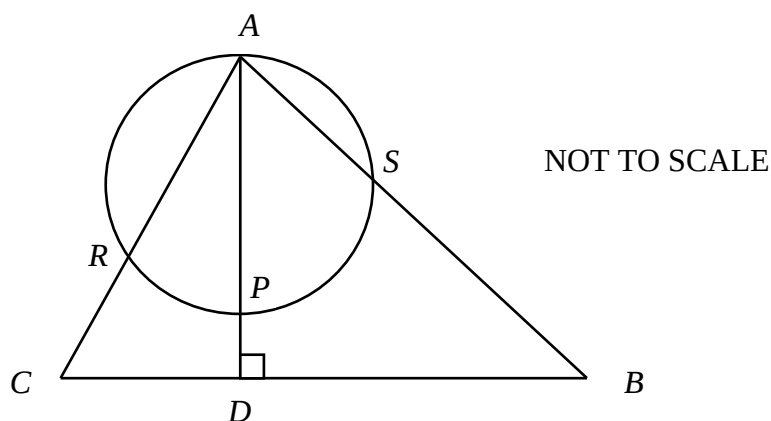
By using the method of cylindrical shells, find the volume of the solid generated.

3

Question 13 continues on page 10

- (c) In the triangle ABC , D is the foot of the altitude from A .

P is any point on AD . The circle drawn with diameter AP cuts AC at R and AB at S .



Prove that $BCRS$ is a cyclic quadrilateral.

3

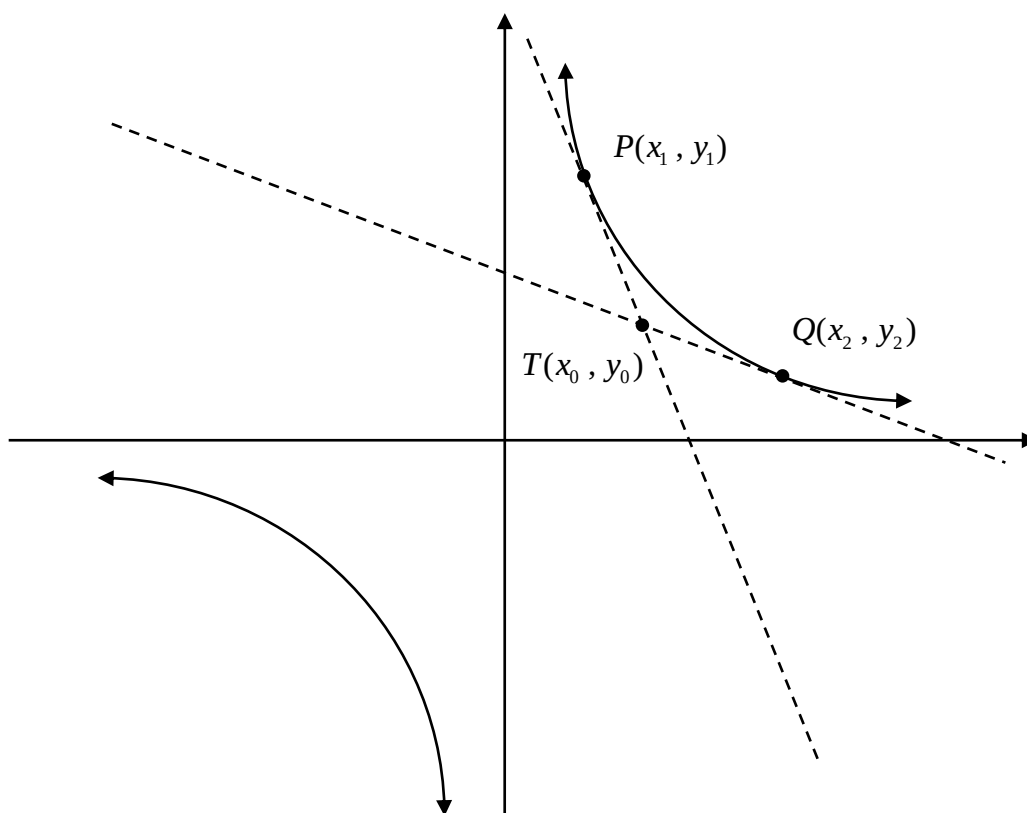
- (d) The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ has asymptotes given by $y = \pm \frac{bx}{a}$.

Show that the product of the lengths of the perpendiculars from any point $P(a \sec \theta, b \tan \theta)$ on the hyperbola to its asymptotes is equal to $\frac{b^2}{e^2}$.

3

End of Question 13

- (a) The tangents at $P(x_1, y_1)$ and $Q(x_2, y_2)$ on the hyperbola $xy = 1$ intersect at $T(x_0, y_0)$.



- (i) Show that the tangent at $P(x_1, y_1)$ has the equation $xy_1 + yx_1 = 2$ 2
- (ii) Show that the chord of contact from T has the equation $xy_0 + yx_0 = 2$ 2
- (iii) Show that x_1 and x_2 are the roots of the equation $y_0x^2 - 2x + x_0 = 0$ 2
- (iv) Hence, or otherwise, show that the midpoint R of PQ has coordinates $\left(\frac{1}{y_0}, \frac{1}{x_0}\right)$. 2
- (v) Hence, or otherwise, show that as T moves on the hyperbola $xy = c^2$, $0 < c < 1$,
 R moves on the hyperbola $xy = \frac{1}{c^2}$. 1

Question 14 continues on page 12

Question 14 continued

(b) A particle is travelling in a straight line with speed v m/s.

Its acceleration is given by the equation $a = -kv$.

Initially it is at the origin travelling at a speed of U m/s.

- | | | |
|-------|---|----------|
| (i) | Prove that $a = v \frac{dv}{dx}$ | 1 |
| (ii) | Find an expression for the particle's velocity v in terms of its displacement x . | 2 |
| (iii) | Find an expression for the particle's velocity v in terms of time t . | 2 |
| (iv) | Hence, or otherwise, find its limiting displacement as $t \rightarrow \infty$ | 1 |

End of Question 14

Question 15 (15 marks)**[Use a SEPARATE writing booklet]****Marks**

- (a) A cubic equation $x^3 + bx^2 + cx + d = 0$ has roots α, β, γ .

α, β, γ are in geometric progression and the middle root $\beta = -6$.

It is also known that $\alpha\beta + \beta\gamma + \alpha\gamma = 156$.

Find the equation of this cubic expressed in the form $x^3 + bx^2 + cx + d = 0$.

3

- (b) It is known that $x^3 - 6x^2 + 11x + a - 4 = 0$ has three distinct integer solutions.

Find the value of a and the three integer solutions.

3

- (c) (i) Find all the fifth roots of 1 and plot them on the unit circle.

1

- (ii) Let ω represent a non-real fifth root of 1, such that $0 < \arg \omega < \frac{\pi}{2}$.

Show that the five roots of 1 can be represented as $1, \omega, \omega^2, \omega^3, \omega^4$

1

- (iii) Show that $(1 - \omega)(1 - \omega^2)(1 - \omega^3)(1 - \omega^4) = 5$

2

- (iv) Show that $(1 - \omega)(1 - \omega^4) = 2 - 2\cos\frac{2\pi}{5}$

2

- (v) Hence, or otherwise, find the exact value of $\sin\frac{\pi}{5}\sin\frac{2\pi}{5}$

3**End of Question 15**

Question 16 (15 marks)**[Use a SEPARATE writing booklet]****Marks**

- (a) When a certain polynomial $P(x)$ is divided by $(x - 2)$ its remainder is -5 .

When $P(x)$ is divided by $(x - 1)$ its remainder is -2 .

Find the remainder when $P(x)$ is divided by $(x - 1)(x - 2)$.

2

- (b) Let $f(x) = \frac{\sin x + 1}{3 \sin x + 2}$.

- (i) Prove that $f(x)$ is a decreasing function for $0 < x < \frac{\pi}{2}$.

1

- (ii) Hence, or otherwise, prove that $\frac{(\sqrt{2} - 1)\pi}{12} < \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sin x + 1}{3 \sin x + 2} dx < \frac{\pi}{28}$

3

- (c) (i) By induction, show that for each positive integer n there are unique positive integers p_n and q_n such that $(1 + \sqrt{2})^n = p_n + q_n \sqrt{2}$

2

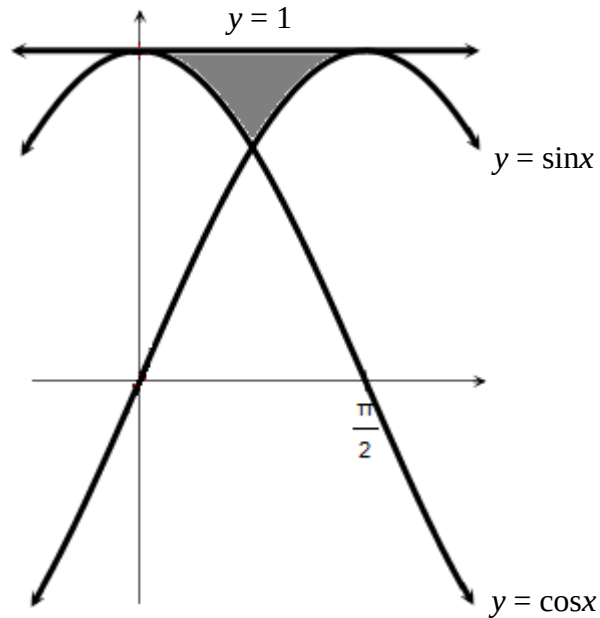
- (ii) Hence also show that $p_n^2 - 2q_n^2 = (-1)^n$

2

Question 16 continues on page 15

- (d) The shaded area in the diagram is bounded by the curves $y = \sin x$, $y = \cos x$ and $y = 1$.

This area is rotated around the y -axis.



- (i) By differentiation or otherwise, show that $\sin^{-1} y + \cos^{-1} y = \frac{\pi}{2}$ **1**
- (ii) By using the method of slicing discs, find the volume of the solid generated. **4**

End of Question 16
End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

(1) Since $b > a$,
 $a^2 = b^2(1 - e^2)$
 $\therefore 4 = 9(1 - e^2)$
 $\therefore 1 - e^2 = \frac{4}{9}$
 $e^2 = \frac{5}{9} \therefore e = \frac{\sqrt{5}}{3} \therefore C$

(2) D

(3) $ae = 3$
 $\frac{a}{e} = \frac{1}{3} \rightarrow 3a = 4e$
 $e = \frac{3a}{4}$

$\therefore a + \frac{3a}{4} = 3$
 $a^2 = 4$
 $a = 2 \rightarrow e = \frac{3}{2}$
 $b^2 = a^2(e^2 - 1)$
 $\therefore b^2 = 4\left(\frac{9}{4} - 1\right)$
 $b^2 = 4 \times \frac{5}{4} = 5$
 $\therefore \frac{x^2}{4} - \frac{y^2}{5} = 1 \therefore A$

(4) $P(2) = P'(2) = 0$
 $\therefore 16 + 4m + 2n + 28 = 0$
 $4m + 2n = -44$
 $2m + n = -22$
 $P'(2) = 4(2)^3 + 2m(2) + n = 0$
 $32 + 2m + n = 0$
 $4m + 2n = -32$
 $\therefore 2m = -10$
 $m = -5$
 $n = -12 \therefore B$

(5) D Since z^2 doubles the argument and the 2 doubles the modulus

(6) Let $y = \frac{1}{x} \therefore x = \frac{1}{y}$
 $\therefore \frac{2x}{y^3} - \frac{5}{y^2} + \frac{2}{y} - 1 = 0$
 $\therefore 2 - 5y + 2y^2 - y^3 = 0$
 $\therefore x^3 - 2x^2 + 5x - 2 = 0 \therefore B$

(7) $2x + 2y \frac{dy}{dx} + y + x \frac{dy}{dx} = 0$
 $\frac{dy}{dx} = \frac{-(2x + y)}{2y + x}$

$\therefore 2x = -y (x \neq 0)$

$\therefore x^2 + 4x^2 - 2x^2 = 5$
 $3x^2 = 5$
 $x = \pm \sqrt{\frac{5}{3}}$

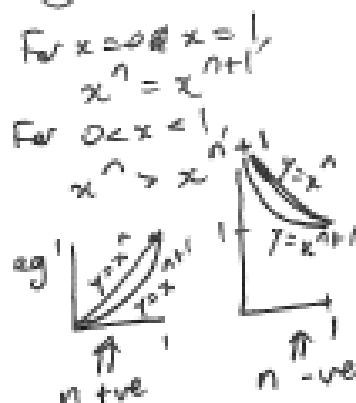
$\therefore$ At points $(\sqrt{\frac{5}{3}}, -2\sqrt{\frac{5}{3}})$ & $(-\sqrt{\frac{5}{3}}, 2\sqrt{\frac{5}{3}})$,
 two distinct horizontal tangents $\therefore C$

(8) Not A since A is 0 due to odd fn
 Not C since $\tan^2 x = \tan x \tan^3 x = \tan x (\sec^2 x - 1)$
 Can't see how B works...
 Maybe... $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ since $\tan \frac{\pi}{4} = 1$

$\therefore \int_0^{\frac{\pi}{4}} (\tan x)^3 dx = \int_0^{\frac{\pi}{4}} \left[\tan\left(\frac{\pi}{4} - x\right) \right]^3 dx$
 $= \int_0^{\frac{\pi}{4}} \left[\frac{\tan \frac{\pi}{4} - \tan x}{1 + \tan \frac{\pi}{4} \tan x} \right]^3 dx$
 $= \int_0^{\frac{\pi}{4}} \left[\frac{1 - \tan x}{1 + \tan x} \right]^3 dx \therefore D$



$= 2 \cdot \left(\frac{2\pi}{3} - \sin \frac{2\pi}{3} \right)$
 $= 2 \cdot \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right)$
 $= \frac{4\pi}{3} - \sqrt{3} \therefore A$



$\therefore \int_0^1 x^n dx > \int_0^1 x^{n+1} dx$
 horizontal tangents $\therefore \int_0^1 1 + x^n dx > \int_0^1 1 + x^{n+1} dx$
 $\therefore \int_0^1 \frac{1}{1+x^n} dx < \int_0^1 \frac{1}{1+x^{n+1}} dx$
 $\therefore A$ is always true

(11) (a) (i) $(a+bi)^2 = 15-8i$
 $\therefore a^2 - b^2 = 15$
 $2ab = -8 \rightarrow b = -\frac{4}{a}$

$\therefore a^2 - \frac{16}{a^2} = 15$
 $a^4 - 15a^2 - 16 = 0$
 $(a^2 - 16)(a^2 + 1) = 0$
 $\therefore a = \pm 4$, since a is real
 $\therefore b = \mp 1$
 $\therefore \sqrt{15-8i} = \pm(4-i)$

(ii) $z = \frac{-6 \pm \sqrt{36 - 4(-6+8i)}}{2}$

$= \frac{-6 \pm \sqrt{36 + 24 - 32i}}{2}$

$= \frac{-6 \pm \sqrt{60 - 32i}}{2}$

$= \frac{-6 \pm 2\sqrt{15-8i}}{2}$

$= -3 \pm \sqrt{15-8i}$

$= -3 \pm (4-i)$

$z = 1-i$ or $-7+i$

(b) (i) $\frac{2\pi}{3}$ $2\cos \frac{\pi}{3}$

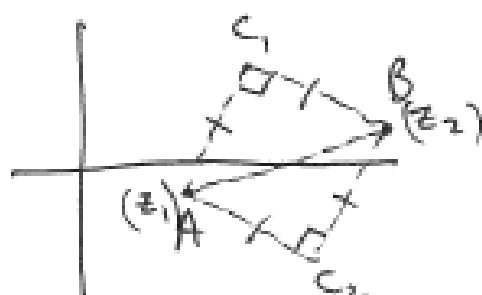
(ii) $2^n \cos \frac{2\pi}{3}$ real?

when $\frac{2\pi}{3} = \pm k\pi$ (k integer)

$\therefore n = \pm 3k$

$\therefore n$ is a multiple of 3

(c)



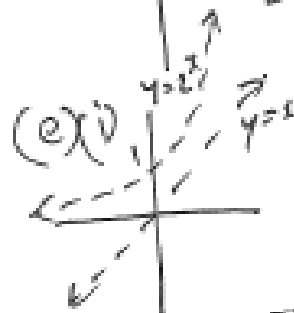
AC, BC is a square
 Midpt $AB = (5, 0)$ which is up 1, across 3 from A
 $M_{AB} = \frac{3}{6} = \frac{1}{2} \therefore M_{C_1, C_2} = -3$

$\therefore$ From $(5, 0)$ go right 1, down 3 for C_2
 & go left 1, up 3 for C_1

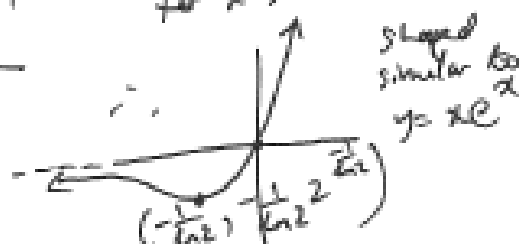
This is because diagonals bisect each other at right angles and are the same length

$\therefore C_2(6, -3) = C_1(4, 3)$

$\therefore z_2 = 6-3i$ & $4+3i$

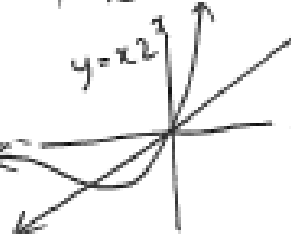


$y = 2^x$ is more powerful for $x \rightarrow -\infty$.



$\frac{dy}{dx} = 1(2^x) + (2^x \ln 2)x$
 $= 2^x(1 + x \ln 2) = 0$ for stat pts
 $\therefore x \ln 2 = -1 \therefore x = -\frac{1}{\ln 2}$

(ii) $x 2^x = kx$



we need $k > 0$, except where $y = kx$ is tangent to $y = x 2^x$ at the origin.

Gradient of $y = x 2^x$ at $x = 0$ is $2^0(1+0) = 1$

$\therefore k > 0$ except $k = 1$

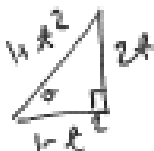
(12) (a) (i) $A(x^2+4) + (x-2)(Bx+C) \equiv 1$
 $x=2: 8A=1 \therefore A=\frac{1}{8}$
 $x=0: 4A-2C=1 \therefore \frac{1}{2}-2C=1$
 $\therefore 2C=-\frac{1}{2}$
 $C=-\frac{1}{4}$

Coeff $x^1: A+B=0 \therefore B=-\frac{1}{8}$

(ii) $\int \frac{\frac{1}{8}}{x-2} + \frac{-\frac{1}{8}x - \frac{1}{4}}{x^2+4} dx$
 $= \frac{1}{8} \int \frac{1}{x-2} - \frac{x+2}{x^2+4} dx$
 $= \frac{1}{8} \int \frac{1}{x-2} - \frac{x}{x^2+4} - \frac{2}{x^2+4} dx$
 $= \frac{1}{8} \int \frac{1}{x-2} - \frac{1}{16} \int \frac{2x}{x^2+4} - \frac{1}{4} \int \frac{1}{x^2+4} dx$
 $= \frac{1}{8} \ln|x-2| - \frac{1}{16} \ln|x^2+4| - \frac{1}{4} \times \frac{1}{2} \tan^{-1} \frac{x}{2} + C$
 $= \frac{1}{8} \ln|x-2| - \frac{1}{16} \ln|x^2+4| - \frac{1}{8} \tan^{-1} \frac{x}{2} + C$

(b) $\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2} = \frac{1}{2} (1 + \tan^2 \frac{\theta}{2}) = \frac{1}{2} (1 + t^2)$
 $\therefore 2dt = (1+t^2) d\theta$
 $\therefore d\theta = \frac{2dt}{1+t^2}$

$\theta=0: t=0$
 $\theta=\frac{\pi}{2}: t = \tan \frac{\pi}{4} = 1$



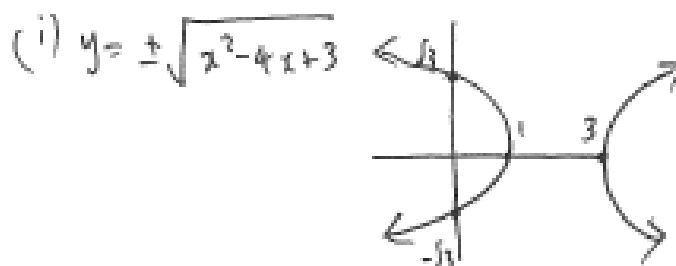
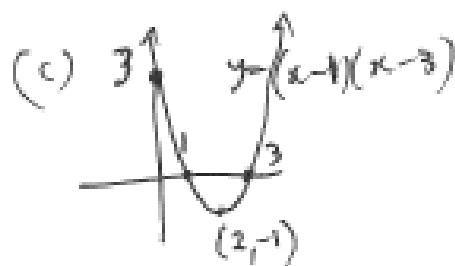
$\therefore \int_0^1 \frac{2dt}{1+t^2} = \frac{1}{\frac{1}{1+t^2} + 2}$

$= \int_0^1 \frac{2dt}{2t^2 + 2 + 2t^2} = \int_0^1 \frac{dt}{t^2 + t + 1}$

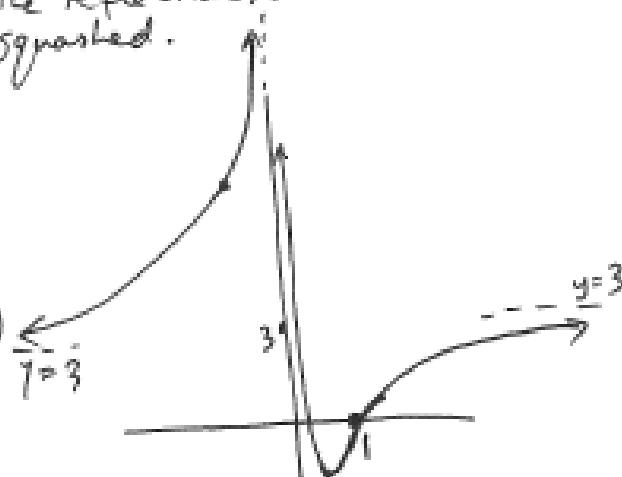
$= \int_0^1 \frac{dt}{(t+\frac{1}{2})^2 + \frac{3}{4}} = \left[\frac{\frac{1}{\sqrt{3}}}{\frac{1}{2}} \tan^{-1} \frac{(t+\frac{1}{2})}{\frac{\sqrt{3}}{2}} \right]_0^1$

$= \left[\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2t+1}{\sqrt{3}} \right) \right]_0^1 = \frac{2}{\sqrt{3}} \left(\tan^{-1} \frac{3}{\sqrt{3}} - \tan^{-1} \frac{1}{\sqrt{3}} \right)$

$= \frac{2}{\sqrt{3}} \left(\tan^{-1} \sqrt{3} - \tan^{-1} \frac{1}{\sqrt{3}} \right) = \frac{2}{\sqrt{3}} \left(\frac{\pi}{3} - \frac{\pi}{6} \right) = \frac{2}{\sqrt{3}} \times \frac{\pi}{6} = \frac{\pi}{3\sqrt{3}} = e(2e-1)$



(ii) $f(x)$ and $f(\frac{1}{x})$ are the same for $x = \pm 1$, and are reflections of each other in the lines $x = \pm 1$, albeit the reflections are stretched and squashed.



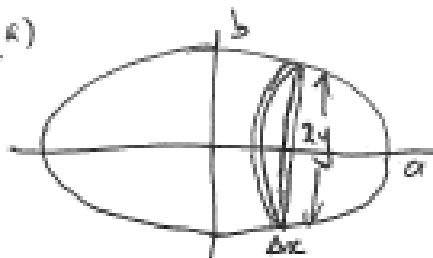
(d) (i) $I_n = \int x^n e^x dx = x^n e^x - \int n x^{n-1} e^x dx$
 $I_n = x^n e^x - n \int x^{n-1} e^x dx$
 $I_n = x^n e^x - n I_{n-1}$

(ii) $I_2 = x^2 e^x - 2I_1$
 $I_1 = x e^x - 2I_0$
 $I_0 = e^x$

$\therefore I_1 = x e^x - 2e^x = e^x (x-2)$
 $\therefore I_2 = x^2 e^x - 2x e^x + 2e^x = e^x (x^2 - 2x + 2)$
 $\therefore I = \left[e^x (x^2 - 2x + 2) \right]_1^2$

$= e^2 (4 - 4 + 2) - e (1 - 2 + 2)$
 $= 2e^2 - e$

(13) (a)



V_1 : cross-section parallel to y-axis

Radius is y

$$\therefore \Delta V_1 = \frac{\pi y^2 \Delta x}{2}$$

$$V_1 = 2 \int_0^a \frac{\pi y^2}{2} dx$$

$$= \pi \int_0^a y^2 dx$$

$$= \pi \int_0^a b^2 \left(1 - \frac{x^2}{a^2}\right) dx$$

$$= \pi b^2 \int_0^a \left(1 - \frac{x^2}{a^2}\right) dx$$

$$= \pi b^2 \left[x - \frac{x^3}{3a^2} \right]_0^a$$

$$= \pi b^2 \left[a - \frac{a}{3} \right] = \frac{2\pi ab^2}{3}$$

V_2 : cross-section parallel to x-axis

Radius is x

$$\therefore \Delta V_2 = \frac{\pi x^2 \Delta y}{2}$$

$$V_2 = 2 \int_0^b \frac{\pi x^2}{2} dy$$

$$= \pi \int_0^b x^2 dy$$

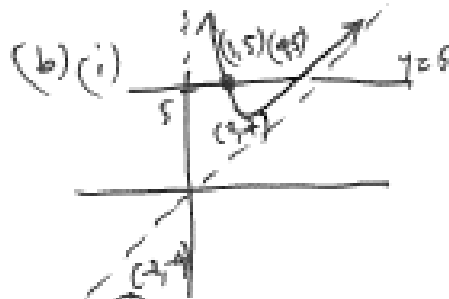
$$= \pi a^2 \int_0^b \left(1 - \frac{y^2}{b^2}\right) dy$$

$$= \pi a^2 \left[y - \frac{y^3}{3b^2} \right]_0^b$$

$$= \pi a^2 \left[b - \frac{b}{3} \right] = \frac{2\pi a^2 b}{3}$$

$$\therefore \frac{V_1}{V_2} = \frac{\frac{2\pi ab^2}{3}}{\frac{2\pi a^2 b}{3}} = \frac{6\pi ab^2}{6\pi a^2 b} = \frac{b}{a}$$

$\therefore$ Ratio $V_1:V_2 = b:a$
or Ratio $V_2:V_1 = a:b$



$$5 = x + \frac{4}{x}$$

$$x^2 - 5x + 4 = 0$$

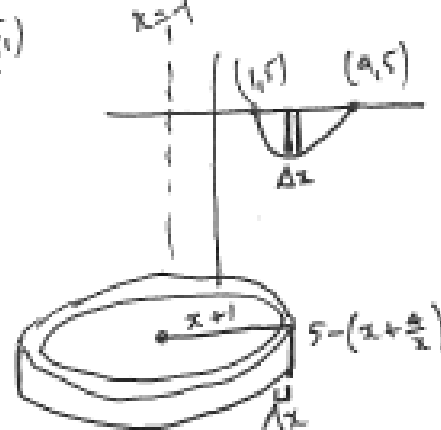
$$(x-1)(x-4) = 0$$

$$x = 1 \text{ or } 4$$

$$g'(x) = 1 - \frac{4}{x^2} = 0 \text{ for stat pts}$$

$$\frac{4}{x^2} = 1 \therefore x^2 = 4 \therefore x = \pm 2$$

(ii)



$$\therefore \Delta V = 2\pi(x+1)\left(5 - x - \frac{1}{x}\right) \Delta x$$

$$\therefore V = 2\pi \int_1^4 \left(x+1\right)\left(5 - x - \frac{1}{x}\right) dx$$

$$V = 2\pi \int_1^4 \left(5x - x^2 - 4 + 5 - x - \frac{1}{x}\right) dx$$

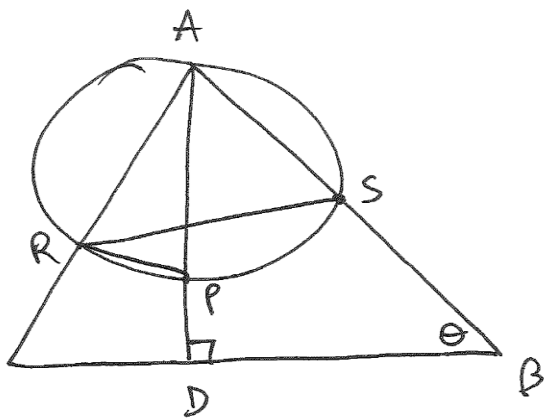
$$= 2\pi \int_1^4 \left(4x - x^2 + 1 - \frac{1}{x}\right) dx$$

$$= 2\pi \left[2x^2 - \frac{x^3}{3} + x - \ln x \right]_1^4$$

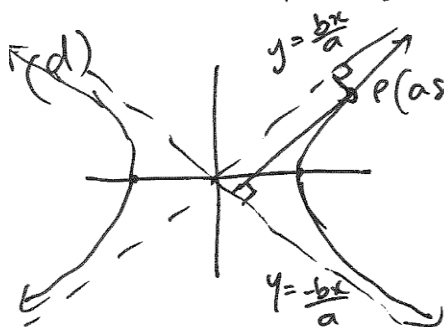
$$= 2\pi \left[32 - \frac{64}{3} + 4 - \ln 4 - 2 + \frac{1}{3} - 1 + 0 \right]$$

$$= 2\pi [12 - 4\ln 4] = 8\pi (3 - \ln 4)$$

(C)



Join RS and RP

let $\angle ABD = \theta$ $\therefore \angle DAB = 90 - \theta$ (angle sum $\triangle ABD$) $\therefore \angle PRS = 90 - \theta$ (angles at circumference on same arc PS) $\angle ARP = 90^\circ$ (angle in semicircle since AP is diameter) $\therefore \angle ARS = 90 - (90 - \theta) = \theta$ (subtraction) $\therefore \angle ARS = \angle ABD$ $\therefore$ BCRS cyclic quad (exterior angle = opposite interior angle) $\therefore$ Product =

$$\frac{|b a \sec \theta - a b \tan \theta|}{\sqrt{b^2 + (-a)^2}} \times \frac{|b a \sec \theta + a b \tan \theta|}{\sqrt{b^2 + a^2}}$$

$$= \frac{|b^2 a^2 \sec^2 \theta + b^2 a^2 \sec \theta \tan \theta - \frac{b^2 a^2 \tan \theta \sec \theta}{-b^2 a^2 \tan^2 \theta}|}{\sqrt{b^2 + a^2} \sqrt{b^2 + a^2}}$$

$$= \frac{|b^2 a^2 (\sec^2 \theta - \tan^2 \theta)|}{b^2 + a^2}$$

$$= \frac{|b^2 a^2 (1 + \tan^2 \theta - \tan^2 \theta)|}{b^2 + a^2}$$

$$= \frac{b^2 a^2}{b^2 + a^2} \text{ since } a, b > 0$$

ie. chord of contact from T is $xy_0 + yx_0 = 2$

$$\text{Now } b^2 = a^2(e^2 - 1)$$

$$\therefore e^2 - 1 = \frac{b^2}{a^2} \therefore e^2 = \frac{b^2}{a^2} + 1 = \frac{b^2 + a^2}{a^2}$$

$$\therefore \frac{1}{e^2} = \frac{a^2}{b^2 + a^2}$$

$$\therefore \frac{b^2 a^2}{b^2 + a^2} = b^2 \times \frac{1}{e^2} = \frac{b^2}{e^2}$$

$$(14) (a) (i) y = \frac{1}{x} \therefore \frac{dy}{dx} = -\frac{1}{x^2}$$

$$\text{At } x = x_1, m = -\frac{1}{x_1^2}$$

$$\therefore y - y_1 = -\frac{1}{x_1^2}(x - x_1)$$

$$\therefore x_1^2 y - x_1^2 y_1 = -x + x_1$$

$$\text{But } x_1 y_1 = 1 \therefore x_1^2 y - x_1 = -x + x_1$$

$$\therefore x_1^2 y + x = 2x_1 \quad (\div x_1)$$

$$\therefore x_1 y + \frac{x}{x_1} = 2$$

$$\text{Now } \frac{1}{x_1} = y_1$$

$$\therefore x_1 y + x y_1 = 2$$

$$\text{ie } x y_1 + y x_1 = 2$$

(ii) Tangent at P passes through T

$$\therefore \text{FACT: } x_0 y_1 + y_0 x_1 = 2$$

Tangent at Q has eqn

$$x y_2 + y x_2 = 2$$

It passes through T

$$\therefore \text{FACT: } x_0 y_2 + y_0 x_2 = 2$$

Consider the line $x y_0 + y x_0 = 2$.Does it pass through P(x₁, y₁)?

$$\text{ie. does } x_1 y_0 + y_0 x_1 = 2?$$

YES!

Does it pass through Q(x₂, y₂)?

$$\text{ie. does } x_2 y_0 + y_0 x_2 = 2?$$

YES!

$$\therefore \text{Since } x y_0 + y x_0 = 2$$

is a line that

passes through

P and Q, the

equation of PQ is

$$x y_0 + y x_0 = 2$$

$$\text{ie. } x y_0 + y x_0 = 2$$

(iii) x_1, x_2 are x -coordinates of simultaneous eqns involving $xy_0 + yx_0 = 2$ & $y = \frac{1}{x}$

$$\therefore xy_0 + \frac{x_0}{x} = 2$$

$$\therefore x^2 y_0 + x_0 = 2x$$

$$\therefore y_0 x^2 - 2x + x_0 = 0$$

(iv) $\frac{x_1 + x_2}{2} = \frac{\frac{2}{y_0}}{2} = \frac{1}{y_0}$

$\therefore x$ coordinate of midpt Pa is $\frac{1}{y_0}$

When $x = \frac{1}{y_0}$, $\frac{1}{y_0} y_0 + yx_0 = 2 \therefore 1 + yx_0 = 2$

$$\therefore yx_0 = 1$$

$$\therefore y = \frac{1}{x_0}$$

$\therefore y$ coordinate of midpt Pa is $\frac{1}{x_0}$

$\therefore P \left(\frac{1}{y_0}, \frac{1}{x_0} \right)$

(v) If $T(x_0, y_0)$ lies on $xy = c^2$ then

$$x_0 y_0 = c^2$$

Does R lie on $xy = \frac{1}{c^2}$?

Subst: Does $\frac{1}{y_0} \times \frac{1}{x_0} = \frac{1}{c^2}$?

Does $\frac{1}{x_0 y_0} = \frac{1}{c^2}$?

Does $\frac{1}{c^2} = \frac{1}{c^2}$? Yes!

$\therefore R$ moves on hyperbola $xy = \frac{1}{c^2}$

(b)(i) $a = \frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt} = \frac{dv}{dx} \cdot v = v \frac{dv}{dx}$

(ii) $-kv = v \frac{dv}{dx} \therefore \frac{dv}{dx} = -k$

$\therefore v = -kx + C$. When $x=0, v=U$

$\therefore U = C \therefore v = U - kx$

(iii) $\frac{dt}{dv} = -\frac{1}{kv}$

$\therefore t = -\frac{1}{k} \ln v + C$

When $t=0, v=U$

$\therefore 0 = -\frac{\ln U}{k} + C \therefore C = \frac{\ln U}{k}$

$\therefore t = \frac{1}{k} \ln \left(\frac{U}{v} \right)$

$\therefore e^{kt} = \frac{U}{v} \therefore v = U e^{-kt}$

(iv) As $t \rightarrow \infty, v \rightarrow 0$ (assuming $k > 0$)

As $v \rightarrow 0, U - kx \rightarrow 0$

ie $U \rightarrow kx$

ie $x \rightarrow \frac{U}{k}$

$\therefore$ As $t \rightarrow \infty, x \rightarrow \frac{U}{k}$

(15) (a) Roots: $-\frac{b}{r}, -b, -br$

Now $\frac{3b}{r} + 3b + 3br = 15b$

$\therefore \frac{3b}{r} + 3br + 3b = 15b$

$9 + 9r + 9r^2 = 39r$

$\therefore 9r^2 - 30r + 9 = 0$

$3r^2 - 10r + 3 = 0$

$(3r-1)(r-3) = 0$

$\therefore r = \frac{1}{3} \text{ or } 3$

$\therefore$ Roots: $-18, -b, -2$

Sum = $-2b$. Product = $-21b$

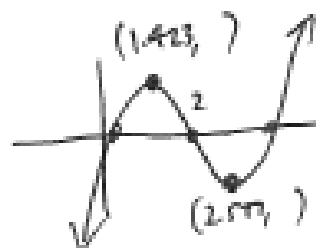
$\therefore x^3 + 2bx^2 + 15bx + 21b = 0$

(b) If $y = x^3 - 6x^2 + 11x + a - 9$,

$\frac{dy}{dx} = 3x^2 - 12x + 11 = 0$ stat pts

$\therefore x = \frac{12 \pm \sqrt{144 - 132}}{6} = \frac{12 \pm \sqrt{12}}{6}$

$= 2 \pm 0.577...$



Since positive cubic with 3 distinct integer solns, 2 stat pts must be above & below x -axis.

Thus $x=2$ must be integer soln

$$\therefore p(2) = 0$$

$$\therefore 8 - 24 + 22 + a - 4 = 0$$

$$\therefore a = -2$$

$$\therefore x^3 - 6x^2 + 11x - 6 = 0$$

$$\therefore (x-2)(x^2 - 4x + 3) = 0$$

$$\therefore (x-2)(x-1)(x-3) = 0$$

$$\therefore a = -2 \text{ \& \# integer solutions: } 1, 2, 3$$

$$\begin{aligned} \text{(iv) LHS} &= 1 - \omega - \omega^4 + \omega^5 \\ &= 1 - (\omega + \omega^4) + 1 \\ &= 2 - (\omega + \omega^4) \end{aligned}$$

$$\begin{aligned} \text{Now } \omega + \omega^4 &= \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + \cos -\frac{2\pi}{5} \\ &= \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} \\ &\quad + \cos -\frac{2\pi}{5} + i \sin -\frac{2\pi}{5} \\ &= 2 \cos \frac{2\pi}{5} \end{aligned}$$

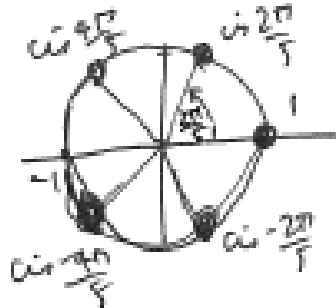
$$\therefore \text{LHS} = 2 - 2 \cos \frac{2\pi}{5}$$

$$\text{(c) (i) } z^5 = 1$$

$$\therefore z^5 = \cos(0 + 2k\pi), \text{ k integer}$$

$$\therefore z = \cos \frac{2k\pi}{5}$$

$$\therefore z = 1, \cos \frac{2\pi}{5}, \cos \frac{4\pi}{5}, \cos -\frac{2\pi}{5}, \cos -\frac{4\pi}{5}$$



$$\begin{aligned} \text{(v) Similarly, } (1 - \omega^2)(1 - \omega^3) &= 2 - (\omega^2 + \omega^3) \\ &= 2 - 2 \cos \frac{4\pi}{5} \end{aligned}$$

$$\therefore (2 - 2 \cos \frac{2\pi}{5})(2 - 2 \cos \frac{4\pi}{5}) = 5$$

$$\therefore (1 - \cos \frac{2\pi}{5})(1 - \cos \frac{4\pi}{5}) = \frac{5}{4}$$

$$\begin{aligned} \text{Now } \cos 2\theta &= 1 - 2 \sin^2 \theta \\ \therefore -\cos 2\theta &= 2 \sin^2 \theta - 1 \\ \therefore 1 - \cos 2\theta &= 2 \sin^2 \theta \end{aligned}$$

$$\therefore (2 \sin^2 \frac{\pi}{5})(2 \sin^2 \frac{2\pi}{5}) = \frac{5}{4}$$

$$\therefore \sin^2 \frac{\pi}{5} \sin^2 \frac{2\pi}{5} = \frac{5}{16}$$

$$\therefore \sin \frac{\pi}{5} \sin \frac{2\pi}{5} = \frac{\sqrt{5}}{4} \left(\begin{array}{l} \text{take +ve} \\ \text{since } \frac{\pi}{5}, \frac{2\pi}{5} \text{ acute} \end{array} \right)$$

$$\text{(16) (a) } p(2) = -5, p(1) = -2$$

$$p(x) = Q(x)(x-1)(x-2) + (ax+b)$$

since remainder is linear
if divisor is quadratic

$$\therefore p(2) = -5 = 0 + 2a + b \therefore 2a + b = -5$$

$$p(1) = -2 = 0 + a + b \therefore a + b = -2$$

$$\therefore a = -3$$

$$b = 1$$

$$\therefore \text{Remainder is } -3x + 1$$

$$\text{(iii) } z^5 = 1 \therefore z^5 - 1 = 0$$

$$\therefore (z-1)(z^4 + z^3 + z^2 + z + 1) = 0$$

But also

$$(z-1)(z-\omega)(z-\omega^2)(z-\omega^3)(z-\omega^4) = 0$$

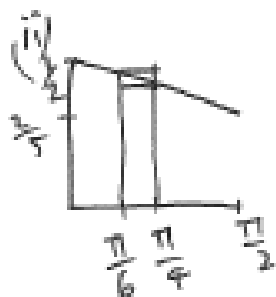
since the 5th roots of 1 are defined in (ii)

$$\therefore (z-1)(z^4 + z^3 + z^2 + z + 1) = (z-1)(z-\omega)(z-\omega^2)(z-\omega^3)(z-\omega^4)$$

$$\therefore z^4 + z^3 + z^2 + z + 1 = (z-\omega)(z-\omega^2)(z-\omega^3)(z-\omega^4)$$

$$\text{Let } z = 1 \therefore 5 = (1-\omega)(1-\omega^2)(1-\omega^3)(1-\omega^4)$$

(b) (i) $f'(x) = \frac{\cos x (3 \sin x + 2) - 3 \cos x (\sin x + 1)}{(3 \sin x + 2)^2}$
 $= \frac{-\cos x}{(3 \sin x + 2)^2}$ For $0 < x < \frac{\pi}{2}$, top -ve, bottom +ve
 $\therefore f'(x) < 0 \therefore$ decreasing



Area squeezed between 2 rectangles,
 width $\frac{\pi}{4} - \frac{\pi}{6} = \frac{\pi}{12}$

$$\therefore \frac{\pi}{12} f\left(\frac{\pi}{4}\right) < \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sin x + 1}{3 \sin x + 2} dx < \frac{\pi}{12} f\left(\frac{\pi}{6}\right)$$

$$\therefore \frac{\pi}{12} \left(\frac{\frac{1}{2} + 1}{\frac{3}{2} + 2} \right) < \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sin x + 1}{3 \sin x + 2} dx < \frac{\pi}{12} \left(\frac{\frac{1}{2} + 1}{\frac{1}{2} + 2} \right)$$

$$\therefore \frac{\pi}{12} \left(\frac{1 + \sqrt{2}}{7 + 2\sqrt{2}} \right) < \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sin x + 1}{3 \sin x + 2} dx < \frac{\pi}{12} \left(\frac{1 + 2}{3 + 4} \right)$$

$$\therefore \frac{\pi}{12} \frac{(1 + \sqrt{2})(3 - 2\sqrt{2})}{9 - 8} < \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sin x + 1}{3 \sin x + 2} dx < \frac{3\pi}{84}$$

$$\therefore \frac{(\sqrt{2} - 1)\pi}{12} < \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sin x + 1}{3 \sin x + 2} dx < \frac{\pi}{28}$$

$\therefore \text{LHS} = (p_n - \sqrt{2}q_n)(p_n + \sqrt{2}q_n)$
 $= (1 - \sqrt{2})^n (1 + \sqrt{2})^n$
 $= ((1 - \sqrt{2})(1 + \sqrt{2}))^n = (-1)^n$
 $= (-1)^n = \text{RHS}$

(d) (i) let $x = \sin^{-1}y + \cos^{-1}y$

$$\therefore \frac{dx}{dy} = \frac{1}{\sqrt{1-y^2}} - \frac{1}{\sqrt{1-y^2}} = 0$$

$\therefore x = \sin^{-1}y + \cos^{-1}y$ always has the same value for all y in $[-1, 1]$

Substitute $y = 1 \therefore x = \sin^{-1}1 + \cos^{-1}1 = \frac{\pi}{2} + 0 = \frac{\pi}{2}$

$$\therefore \sin^{-1}y + \cos^{-1}y = \frac{\pi}{2}$$



Now $\sin x = \cos x$ has soln $x = \frac{\pi}{4}$
 $\therefore \sin \frac{\pi}{4} = \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$

$$\therefore \Delta V = \pi ((\sin^{-1}y)^2 - (\cos^{-1}y)^2) dy$$

$$\therefore V = \int_{\frac{1}{\sqrt{2}}}^1 \pi (\sin^{-1}y - \cos^{-1}y)(\sin^{-1}y + \cos^{-1}y) dy$$

$$= \pi \int_{\frac{1}{\sqrt{2}}}^1 (\sin^{-1}y - \cos^{-1}y) dy$$

$$= \pi \int_{\frac{1}{\sqrt{2}}}^1 \sin^{-1}y - \left(\frac{\pi}{2} - \sin^{-1}y \right) dy$$

$$= \pi \int_{\frac{1}{\sqrt{2}}}^1 2 \sin^{-1}y - \frac{\pi}{2} dy$$

$$= \pi \int_{\frac{1}{\sqrt{2}}}^1 \sin^{-1}y dy - \frac{\pi^2}{4} \int_{\frac{1}{\sqrt{2}}}^1 1 dy$$

$$= \pi^2 \left[(y \sin^{-1}y) - \int \frac{1}{\sqrt{1-y^2}} dy \right]_{\frac{1}{\sqrt{2}}}^1 - \frac{\pi^2}{4} \left[y \right]_{\frac{1}{\sqrt{2}}}^1$$

$$= \pi^2 \left[\frac{\pi}{2} - \frac{1}{\sqrt{2}} \frac{\pi}{4} \right] - \frac{\pi^2}{2} \int_{\frac{1}{\sqrt{2}}}^1 \frac{2y dy}{\sqrt{1-y^2}} - \frac{\pi^2}{4} \left[1 - \frac{1}{\sqrt{2}} \right]$$

$$= \frac{\pi^3}{2} - \frac{\pi^3}{4\sqrt{2}} + \left[\frac{\pi^2}{2} \sqrt{1-y^2} \right]_{\frac{1}{\sqrt{2}}}^1 - \frac{\pi^3}{4} + \frac{\pi^3}{4\sqrt{2}}$$

$$= \frac{\pi^3}{2} - \frac{\pi^3}{4\sqrt{2}} - \pi^2 \sqrt{\frac{1}{2}} - \frac{\pi^3}{4} + \frac{\pi^3}{4\sqrt{2}}$$

$$= \frac{\pi^3}{4} - \frac{\pi^2}{\sqrt{2}}$$

(c) (i) let $n=1$. LHS $= 1 + \sqrt{2} \therefore p_1 = 1, q_1 = 1 \therefore$ True for $n=1$

let $n=k$ be an integer such that $(1 + \sqrt{2})^k = p_k + q_k \sqrt{2}$
 where p_k, q_k are unique integers

Prove $n=k+1$ is an integer such that $(1 + \sqrt{2})^{k+1} = p_{k+1} + q_{k+1} \sqrt{2}$
 where p_{k+1}, q_{k+1} are unique integers

Proof: LHS $= (1 + \sqrt{2})^k (1 + \sqrt{2}) = (p_k + q_k \sqrt{2})(1 + \sqrt{2})$
 $= p_k + 2q_k + \sqrt{2}(p_k + q_k)$

Since p_k, q_k are unique integers,

so are $p_k + 2q_k$ & $p_k + q_k$

$$\therefore = p_{k+1} + q_{k+1} \sqrt{2}$$

$\therefore$ True by mathematical induction

(ii) Now prove $p_n - q_n \sqrt{2} = (1 - \sqrt{2})^n$, with p_n, q_n same values found in (i)

For $n=1$, RHS $= 1 - \sqrt{2}$, LHS $= p_1 - q_1 \sqrt{2} \therefore p_1 = 1, q_1 = 1$ which are same as in (i) $\therefore$ True for $n=1$

let $n=k$ be integer such that $(1 - \sqrt{2})^k = p_k - q_k \sqrt{2}$, p_k, q_k as in (i)

Prove $n=k+1$ is an integer such that $(1 - \sqrt{2})^{k+1} = p_{k+1} - q_{k+1} \sqrt{2}$, p_{k+1}, q_{k+1} as in (i)

Proof: LHS $= (1 - \sqrt{2})(p_k - q_k \sqrt{2}) = p_k + 2q_k - \sqrt{2}(p_k + q_k) = p_{k+1} - q_{k+1} \sqrt{2}$ which are same as in (i)

$\therefore$ True by mathematical induction